



CSA1402 COMPILER DESIGN

INTRODUCTION TO COMPILERS

PRESENTED BY:

AKSHAYA A

192421222

COMPILER

A compiler is a software program.

It converts **high-level language** into **machine language**.

It checks the program for errors. It produces an executable file.

translates a program written in a **high-level programming language** (such as C, C++, or Java) into **machine language**,

IMPORTANCE OF COMPILERS

Makes programs understandable to the computer.

Detects syntax and semantic errors. Improves execution speed.

Produces optimized machine code,saves programming time.

Since programmers write programs using high-level languages, a compiler is required to translate them into machine language.

Working of a Compiler

The programmer writes the source code.

The compiler translates it into machine code.

If errors are found, they are displayed.

If there are no errors, an executable file is generated.

The computer runs the executable and displays the output.

- Source Program
- ↓
- Compiler
- ↓
- Machine Code
- ↓
- Computer Executes
- ↓
- Output

Features of a Compiler

- Converts high-level language into machine language.
- Checks syntax and semantic errors.
- Generates optimized code for better performance.
- Produces executable files.
- Improves program efficiency.
- Supports debugging by displaying error messages.

Advantages of a Compiler

- Faster program execution.
- Detects errors before running the program.
- Optimizes the generated machine code.
- Reduces execution time.
- Improves software performance.
- Suitable for developing large applications.

Applications of a Compiler

- C Programming (GCC Compiler)
- C++ Programming
- Java Programming (javac)
- Operating Systems
- Embedded Systems
- Mobile Application Development
- Game Development
- Software Engineering

CONCLUSION

- A compiler is an essential software tool in computer programming.
- It converts high-level language into machine language, checks for errors, and generates optimized executable code.
- Without a compiler, computers cannot execute programs written by programmers.
- Therefore, compilers play an important role in software development.



THANK YOU

